Tatkal Booking System

Containerization and Deployment of the Microservices

Scalable Services (SEZG583) - Assignment 2 - Deployment Report

Group member	BITS ID	Main work in Assignment 2
Dhruv Bhat	2025TM93139	`docker-compose.yml`, networks and secrets, restart check, report
Ratani Snehi Rajesh Darshana	2025TM93530	Dockerfiles and images, health endpoints
Atharva Kadam	2025TM93221	the deployment problem (Task 3), configuration and start-up logging
N Abhishek Reddy	2023MT93400	Kubernetes manifests, scaling and rollout, demo video

Source code: https://github.com/spaceman-dev/tatkal-booking-microservices (Assignment 1 ends at the tag `assignment-1`, everything after it is Assignment 2)
Demo video: [video link]

1. Application and deployment overview

The application is our Assignment 1 project, the back end for booking Indian Railways Tatkal tickets: a few seats open at a fixed time and thousands of people try to book them in the same minute. It has four business services and a mock bank, all in Python (FastAPI). booking-service runs each booking as a saga: hold seats in inventory-service, take the payment through payment-service (which talks to the bank), confirm the seats, and tell notification-service through an outbox. Each business service owns its own PostgreSQL database, and the services talk to each other only over REST.

For this assignment we built one image per service and run the whole system in two ways: with Docker Compose (`docker compose up --build`, nine containers) and on Minikube. The code does not know where it runs. Every address and password reaches it through environment variables. Figure 1 shows the Compose deployment.

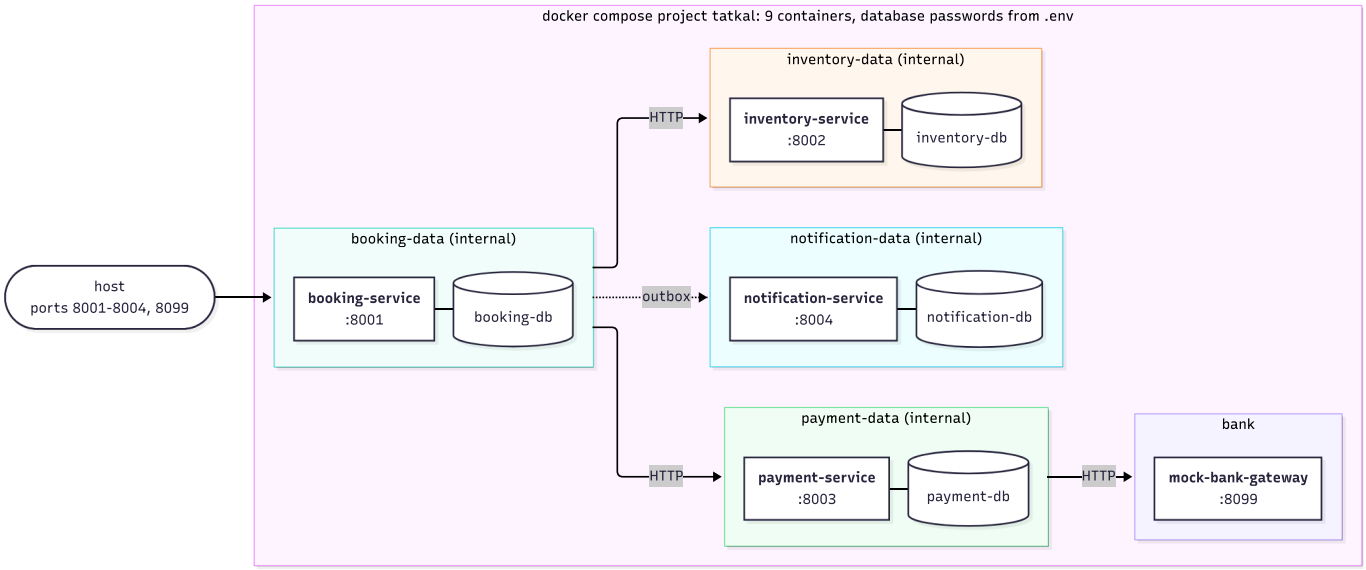


Figure 1 - Docker Compose deployment. Each database is on an internal network shared only with its own service, so booking-service cannot even resolve inventory-db. The services call each other by name on the network `services` (http://inventory-service:8002), and only payment-service is on the network `bank`. Ports 8001-8004 and 8099 are published for the tests, the databases are not. Each database keeps its data in a named volume.

2. Containerization and Docker Compose

2.1 Images

Each service has a Dockerfile in its own folder (`services/<name>/Dockerfile`), built as `tatkal/<name>:1.0`. All five follow the same pattern:

- **Base and layers.** `python:3.12-slim`. `requirements.txt` is copied and installed before the code, so a code change rebuilds only the last layers. The four services with a database come to 348 MB each, the mock bank to 274 MB. pytest moved to `requirements-dev.txt` so it no longer goes into the images.
- **Non-root.** The code is owned by root and the process runs as `tatkal` (uid 10001), so the service can read its code but not change it.
- **Configuration from outside.** Only `HOST`, `PORT` and `APP_VERSION` have defaults in the image. The database URL, the other services' URLs and the log level are given when the container starts. `.dockerignore` keeps `.env`, `.venv` and the tests out of the build context, so a password cannot be copied in by mistake. We checked the built images with `docker image inspect` and by listing `/app` inside them.
- **Health check.** `HEALTHCHECK` calls the service's own `/health` with Python (the slim image has no curl). `--start-interval=2s` makes a new container healthy a few seconds after start-up.
- **Clean stop.** `CMD ["python", "-m", "app"]` in exec form, so Python is PID 1 and receives SIGTERM from `docker stop` and from Kubernetes.

We also added to the code: `/health` reports the version and the container (`instance`), and a new `/health/live` only says the process is running, without touching the database. Compose and the Kubernetes readiness probe use `/health`, the liveness probe uses `/health/live`. That way a database outage takes a pod out of its Service instead of restarting it over and over.

2.2 docker-compose.yml

- **Nine containers:** four `postgres:16-alpine` databases, each with its own user, database and volume, and the five services.
- **Passwords only in `.env`.** The file uses `${BOOKING_DB_PASSWORD:?...}`, so Compose stops with a clear message when `.env` is missing, and builds each `DATABASE_URL` from it. `.env` is ignored by git; the repository has `.env.example` with placeholders.
- **Service names, never localhost:** `INVENTORY_SERVICE_URL=http://inventory-service:8002`, and the database host is `booking-db:5432`.
- **Six networks:** `services` for the REST calls, `bank` for payment-service and the bank only, and one `internal` network per database, shared with its own service only.
- **Start order:** `depends_on` with `condition: service_healthy`. The databases come up first, then each service once its database is ready, and booking-service last because it calls the other three.
- `restart: unless-stopped` everywhere, and json-file logs capped at 3 x 10 MB per container.

2.3 Running it

What we did	Result (files in `evidence/deployment/`)
`docker compose build --no-cache`	five images built with BuildKit (`1-images`)
`docker compose up --build -d --wait`	all nine containers healthy about 11 s after the start (`2-compose/up.txt`)
`/health` of each service, `docker inspect`	all `UP` with the database `UP`, 0 restarts (`health.txt`)
main workflow: 7 end-to-end tests and the demo script	book, pay, confirm, notify; a failure at step 3 refunded and released; a declined payment; sold out. 7 passed (`e2e.txt`, `demo_flow.txt`)
one booking followed through the logs	its correlation id appears in all five containers, in order (`trace.txt`)
name lookups from inside booking-service	its own database and the three services resolve, the other databases and the bank do not (`networks.txt`)
`docker compose down`, then `up -d --wait`	the newest confirmed booking (PNR 8394445298) is still there, the 7 tests pass again (`restart.txt`)

3. Situated learning: the container that could not find its database

Problem. Once the first Dockerfile was ready we ran the inventory-service image on its own, with the `.env` file we had been using on the laptop since Assignment 1: `docker run -d --name inventory-smoke -p 8002:8002 --env-file services/inventory-service/.env tatkal/inventory-service:1.0`. The development database was up and healthy on `localhost:5442`, and the service with that same file had worked fine outside Docker. The container started but stayed at `health: starting`, curl got an empty reply, and after 30 seconds it exited with code 3. The log said `database not reachable yet (attempt n/30)` thirty times and then `Application startup failed`.

Diagnosis. We worked from the outside in:

Command	What it told us
`docker ps -a`, `docker inspect -f '{{.State.ExitCode}}'`	exit code 3 and unhealthy: the service gave up on its own, nothing killed it
`docker logs`	30 failed attempts one second apart, but no reason
`docker inspect` (environment, networks)	`DATABASE_URL=...@localhost:5442/inventory_db`; the container is on the default `bridge` network
`docker compose -f docker-compose.db.yml ps`, `docker inspect tatkal-inventory-db`	the database is healthy, published as 5442 on the host, and on the network `tatkal-db_default` with the DNS name `inventory-db`
`docker exec ... grep localhost /etc/hosts`	inside the container `localhost` is `127.0.0.1` / `::1` of the container itself
`docker exec ... getent hosts inventory-db`	the name does not resolve from the `bridge` network
a database connection from a container of the same image	`connection to server at "127.0.0.1", port 5442 failed: Connection refused`

The image was fine, the configuration was wrong. `localhost:5442` means “this machine”, and inside a container this machine is the container. Port 5442 exists only on the host, where Docker publishes the database. To confirm it we changed only the address in the URL (`localhost:5442` to `inventory-db:5432`) and attached the database’s network: the same image with the same file was healthy within 8 seconds, with no failed attempt.

Solution. (1) In `docker-compose.yml` every `DATABASE_URL` uses the database’s service name and port 5432, and each service shares a private network with its database. The other services’ URLs work the same way. (2) The per-service `.env` files are only for running on the host. They stay out of the images (`.dockerignore`) and are never passed to containers, which get their settings from Compose, and on Kubernetes from ConfigMaps. (3) The wait-for-database log line now shows the target, with the password hidden, and the reason, so the cause is in the first line: `...@localhost:5442/inventory_db ... Connection refused`.

Validation. In the Compose deployment, `docker compose exec inventory-service printenv DATABASE_URL` shows `inventory-db:5432`, `getent hosts inventory-db` resolves, `/health` reports the database `UP` and the 7 end-to-end tests pass. `evidence/deployment/3-troubleshooting` has each step with screenshots, plus the raw output from the morning it first happened.

Learning.

- Inside a container, `localhost` is the container. Containers reach each other by name on a shared network. A published port is for the host, not for other containers.
- Configuration does not carry over from a laptop into containers. Each environment needs its own configuration (Compose `environment`, ConfigMaps), and the developer’s `.env` should never go in.
- A log line that only says “not reachable” made the diagnosis slow. Logging the target and the reason turns minutes into seconds.
- Run one container alone before wiring up the rest. When it fails, only the image or its configuration can be wrong.

In a real deployment we would apply this by making every service log its effective configuration (without secrets) at start-up and fail fast with a clear message when a dependency is missing, and by keeping each environment’s configuration in its deployment files rather than in files copied from someone’s machine.

4. Advanced: Kubernetes on Minikube

4.1 Deployment and main workflow

`deploy/k8s/` has the manifests (Figure 2), and `bash deploy/k8s/deploy.sh` deploys everything to Minikube 1.39 (Kubernetes 1.37):

- It builds the images with Compose and copies them in with `minikube image load` (no registry, `imagePullPolicy: IfNotPresent`), creates the Secret `db-passwords` from the same `.env`, and runs `kubectl apply -k`.
- Each database is a StatefulSet with a headless Service and a 1 Gi PersistentVolumeClaim. Each service is a Deployment, a ClusterIP Service and a ConfigMap with the other services' URLs. `DATABASE_URL` is put together from the Secret with `$(DB_PASSWORD)`.
- An init container waits with `pg_isready` until the pod's database answers, like `depends_on` in Compose. Startup and liveness probes use `/health/live`, the readiness probe `/health`. The service and database containers have CPU and memory requests and a memory limit.
- booking-service runs 2 replicas with `maxSurge: 1, maxUnavailable: 0`. inventory-service and the mock bank stay at one replica, because the fault switches used in the demo are kept in memory.

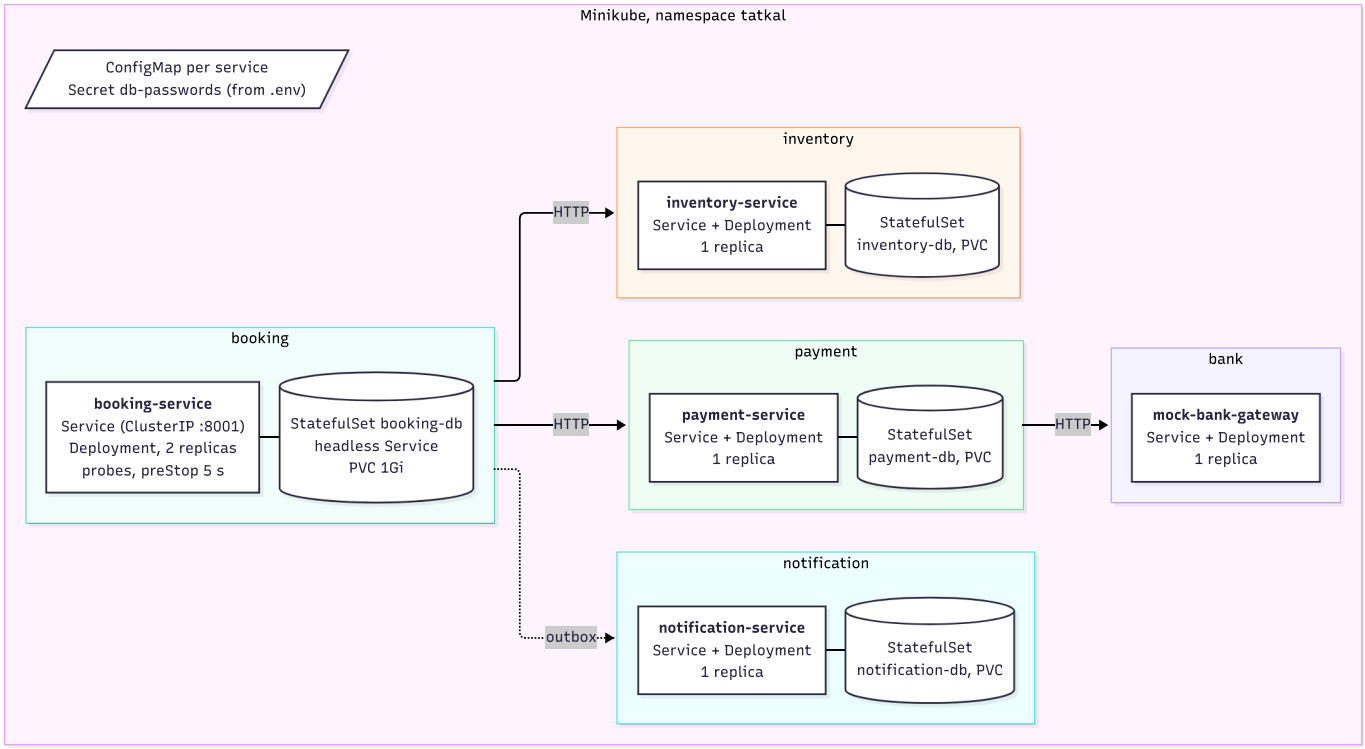


Figure 2 - The Minikube deployment (namespace tatkal). Each service has a Service and a Deployment, and each database a StatefulSet with a headless Service and a volume, as shown for booking-service. The tests reach the cluster through kubectl port-forward on the same local ports as with Compose.

A clean deploy took 41 s. The very first one took 3 min 36 s, almost all of it waiting for the pods while Minikube downloaded the postgres image. The main workflow runs through the Kubernetes Services: with `port-forward.sh` running, the same 7 end-to-end tests pass, reading the logs with `kubectl logs`, and one booking's correlation id appears in the logs of all five Deployments.

4.2 Self-healing and scaling

`demo-scaling.sh` deletes one of the two booking-service pods and then scales the Deployment to 3, while a pod inside the cluster calls booking-service twice a second. A replacement pod was running 3 s after the delete, and the third replica was ready 3 s after `kubectl scale`. The client made 80 requests, all answered, by four different pods.

4.3 Rolling update and rollback

`demo-rollout.sh` rolls booking-service out to 1.1 (the same code with `APP_VERSION=1.1.0`), rolls back with `kubectl rollout undo`, then deploys a tag that does not exist (9.9). That pod never gets ready, `maxUnavailable: 0` keeps the three 1.0 pods serving, and a second `undo` removes it. 159 requests during all of this, none failed.

It was not like that the first time. 3 of 141 requests failed during our first update and rollback, each just as an old pod was stopping: Kubernetes sends SIGTERM and removes the pod from the Service endpoints at the same time, so for a moment requests still reached a stopping pod. A `preStop` sleep of 5 s fixed it (0 of 139 afterwards). Running replicas also showed a race we had never seen with one copy of each service: replicas that create their tables at the same moment can crash with `UniqueViolation` (3 of 4 in our test). Table creation now runs under a PostgreSQL advisory lock. Both are described in Appendix C.

5. Contributions and AI use

Member	Assignment 2 work	Share
Dhruv Bhat	`docker-compose.yml` (networks, health-based start order, passwords from `.env`), `restart_check.sh`, the evidence runs, report sections 1, 2.2 and 5	25 %
Ratani Snehi Rajesh Darshana	the five Dockerfiles and `.dockerignore`, `/health` with version and instance, `/health/live`, checking the images, report section 2.1	25 %
Atharva Kadam	Task 3: reproducing and diagnosing the localhost problem, the better start-up log, `.env.example` files with placeholders, report section 3	25 %
N Abhishek Reddy	Kubernetes manifests and scripts, the preStop and advisory-lock fixes, report section 4, video editing	25 %

AI-use declaration. We used GitHub Copilot in VS Code (agent mode) for this assignment too. It wrote the Dockerfiles, `docker-compose.yml`, the Kubernetes manifests, the deploy and demo scripts and the small code changes (health endpoints, start-up log, `create_tables`). It ran the builds and deployments and recorded the evidence. The problems in section 3 and Appendix C came up during those runs, and the files in `evidence/` are not edited (passwords are masked when they are printed). It also drafted this report, the READMEs and a plan for the video. Our part: we decided what to show for Task 3 and which advanced tasks to attempt, went through every Dockerfile, the Compose file and the manifests until each of us could explain them, and each of us ran `docker compose up --build` from a fresh clone on our own laptop.

Appendix A - Commands

```
# Docker Compose
cp .env.example .env                          # put your own passwords in .env
docker compose up --build -d --wait
docker compose ps                             # 9 containers, all healthy
python3 -m venv .venv && .venv/bin/pip install -r scripts/requirements.txt    # Python 3.11+, tests only
.venv/bin/python -m pytest tests/e2e -v
bash scripts/demo_flow.sh
bash scripts/restart_check.sh

# the Task 3 problem, on purpose (the .env.example has the same localhost URL as our old .env)
docker compose -f docker-compose.db.yml up -d --wait inventory-db
docker run -d --name inventory-host-env -p 8012:8002 \
  --env-file services/inventory-service/.env.example tatkal/inventory-service:1.0
docker ps -a; docker logs inventory-host-env 2>&1 | tail -3
docker inspect -f '{{range .Config.Env}}{{println .}}{{end}}' inventory-host-env | grep DATABASE_URL
docker exec inventory-host-env getent hosts inventory-db; echo $?
docker rm -f inventory-host-env; docker compose -f docker-compose.db.yml down

# Kubernetes
docker compose down                           # frees the local ports for the port-forward
minikube start --cpus=2 --memory=3500mb
bash deploy/k8s/deploy.sh
bash deploy/k8s/port-forward.sh               # second terminal
LOG_SOURCE=k8s .venv/bin/python -m pytest tests/e2e -v
bash deploy/k8s/demo-scaling.sh
bash deploy/k8s/demo-rollout.sh
```

Appendix B - Evidence index

Everything was recorded on 29 Oct 2026. Every file starts with the command, the time and the commit.

Folder in `evidence/deployment/`	What it shows
`1-images`	the clean build of the five images; sizes, user, ports, command, environment and files of each image
`2-compose`	`up`, health, the 7 end-to-end tests, the demo script, a trace across services, network isolation, logs, `down` and `up` with the data kept, and our very first `compose up`
`3-troubleshooting`	the Task 3 problem: symptom, exit, diagnosis, the one-change confirmation, the fix and the tests; the raw output of the first time it happened
`4-kubernetes`	deploy, resources, the 7 tests through port-forward, demo, trace, self-healing and scaling (80 requests, 0 failed), rollout and rollback (159 requests, 0 failed), and the first deploy
`5-other-problems`	the rolling update that dropped requests, before and after the preStop fix; the table-creation race before and after the advisory lock

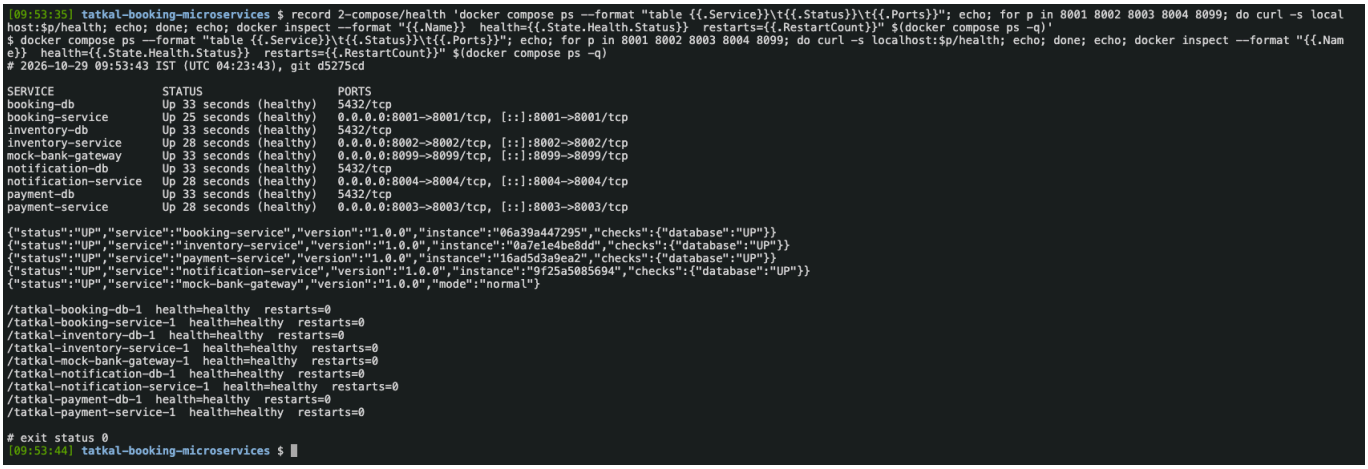
The service test suites were run again after the changes: 133 passed (`evidence/tests`).

Appendix C - Other problems we hit

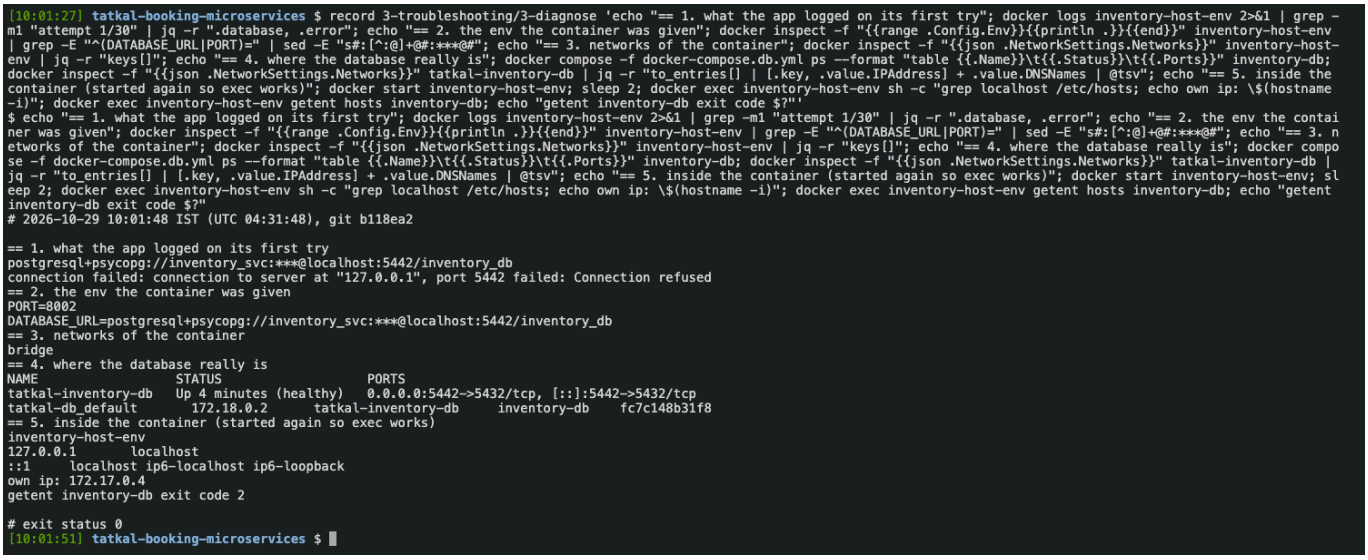
Problem	How it showed up	What we changed
`docker compose` was not found on the Mac	`docker: unknown command: docker compose`, although the plugin was installed with Homebrew	pointed Docker at Homebrew's plugin folder (`cliPluginsExtraDirs` in `~/.docker/config.json`)
the first build used the old builder	Compose warned that it `requires buildx plugin` (`2-compose/first-up.txt`)	installed buildx; builds use BuildKit since

Problem	How it showed up	What we changed
one end-to-end test failed against the containers	6 of 7 passed: the log check read old files in `logs/`, while the containers log to stdout	the test and `trace.py` read `docker compose logs` or `kubectl logs` (`LOG_SOURCE`, detected automatically)
`demo_flow.sh` failed at step 1	`python3` on macOS is 3.9, the scripts need 3.11+	the script uses the virtualenv's Python; `stack.py` checks the version and says what to install
a rolling update dropped requests	3 of 141 failed (connection refused, timeouts) while old pods stopped	`preStop` sleep 5 s and `terminationGracePeriodSeconds: 30`: 0 of 139, later 0 of 159 (commit `c264040`)
replicas starting together crashed	`race_check.py` with 4 replicas on an empty schema: 3 `UniqueViolation` on `pg_type_typname_nsp_index`	`create_tables()` takes `pg_advisory_xact_lock` first; 4 of 4 start; regression test added (commit `4511fed`)

Appendix D - Screenshots



Compose: nine containers healthy, every service UP



Task 3 diagnosis: the target and reason from the log, the environment, the networks, and localhost inside the container

```
[10:03:09] tatkal-booking-microservices $ record 3-troubleshooting/5-fixed 'echo "== the fix is in docker-compose.yml: db by service name, on a shared network, password from
.env"; sed -n "/^  inventory-service:/,/^$/p" docker-compose.yml; echo "== the dev database from the host setup is not needed"; docker compose -f docker-compose.db.yml stop i
nventory-db; echo "== the compose container"; docker compose ps --format "{{.Name}}\t{{.Status}}\t{{.Networks}}" inventory-service inventory-db; docker compose exec inventory
-service printenv DATABASE_URL | sed -E "s#:[^:@]+@#:***@#"; docker compose exec inventory-service getent hosts inventory-db; curl -s localhost:8002/health; echo; echo "== th
e main flow end to end"; .venv/bin/python -m pytest tests/e2e -q'
$ echo "== the fix is in docker-compose.yml: db by service name, on a shared network, password from .env"; sed -n "/^  inventory-service:/,/^$/p" docker-compose.yml; echo "==
 the dev database from the host setup is not needed"; docker compose -f docker-compose.db.yml stop inventory-db; echo "== the compose container"; docker compose ps --format "
{{.Name}}\t{{.Status}}\t{{.Networks}}" inventory-service inventory-db; docker compose exec inventory-service printenv DATABASE_URL | sed -E "s#:[^:@]+@#:***@#"; docker compos
e exec inventory-service getent hosts inventory-db; curl -s localhost:8002/health; echo; echo "== the main flow end to end"; .venv/bin/python -m pytest tests/e2e -q
# 2026-10-29 10:03:24 IST (UTC 04:33:24), git b118ea2

== the fix is in docker-compose.yml: db by service name, on a shared network, password from .env
  inventory-service:
    <<: *service
    build: services/inventory-service
    image: tatkal/inventory-service:${IMAGE_TAG:-1.0}
    environment:
      DATABASE_URL: postgresql+psycopg://inventory_svc:${INVENTORY_DB_PASSWORD}@inventory-db:5432/inventory_db
      ADMIN_API_ENABLED: "true"   # dev/demo only: fault injection used by the tests and the demo
    ports: ["8002:8002"]
    networks: [services, inventory-data]
    depends_on:
      inventory-db: {condition: service_healthy}

== the dev database from the host setup is not needed
 Container tatkal-inventory-db Stopping
 Container tatkal-inventory-db Stopped
== the compose container
tatkal-inventory-db-1   Up 6 minutes (healthy)  tatkal_inventory-data
tatkal-inventory-service-1      Up 6 minutes (healthy)  tatkal_inventory-data,tatkal_services
postgresql+psycopg://inventory_svc:***@inventory-db:5432/inventory_db
172.20.0.2      inventory-db
{"status":"UP","service":"inventory-service","version":"1.0.0","instance":"48c467598142","checks":{"database":"UP"}}
== the main flow end to end
.......                                                                  [100%]
7 passed in 6.76s

# exit status 0
[10:03:32] tatkal-booking-microservices $
```

Task 3 after the fix: the service name, the shared network, health UP, 7 end-to-end tests passed

```
[10:07:53] tatkal-booking-microservices $ record 4-kubernetes/scaling 'bash deploy/k8s/demo-scaling.sh'
$ bash deploy/k8s/demo-scaling.sh
# 2026-10-29 10:07:57 IST (UTC 04:37:57), git b118ea2

== booking-service pods
NAME                               READY   STATUS    RESTARTS   AGE
booking-service-654dbd4fcd-4jppl   1/1     Running   0          2m17s
booking-service-654dbd4fcd-7z8gl   1/1     Running   0          2m17s

== kubectl delete pod booking-service-654dbd4fcd-4jppl
pod "booking-service-654dbd4fcd-4jppl" deleted from tatkal namespace
NAME                               READY   STATUS        RESTARTS   AGE
booking-service-654dbd4fcd-4jppl   1/1     Terminating   0          2m20s
booking-service-654dbd4fcd-4qcjv   0/1     Running       0          3s
booking-service-654dbd4fcd-7z8gl   1/1     Running       0          2m20s
Waiting for deployment "booking-service" rollout to finish: 1 of 2 updated replicas are available...
deployment "booking-service" successfully rolled out
NAME                               READY   STATUS        RESTARTS   AGE
booking-service-654dbd4fcd-4jppl   1/1     Terminating   0          2m20s
booking-service-654dbd4fcd-4qcjv   1/1     Running       0          3s
booking-service-654dbd4fcd-7z8gl   1/1     Running       0          2m20s

== kubectl scale deployment booking-service --replicas=3
deployment.apps/booking-service scaled
Waiting for deployment "booking-service" rollout to finish: 2 of 3 updated replicas are available...
Waiting for deployment "booking-service" rollout to finish: 2 of 3 updated replicas are available...
deployment "booking-service" successfully rolled out
NAME                               READY   STATUS      RESTARTS   AGE
booking-service-654dbd4fcd-4jppl   0/1     Completed   0          2m23s
booking-service-654dbd4fcd-4qcjv   1/1     Running     0          6s
booking-service-654dbd4fcd-7z8gl   1/1     Running     0          2m23s
booking-service-654dbd4fcd-zlhxd   1/1     Running     0          3s

== what the client inside the cluster saw: 80 requests, answers per pod
   6 booking-service-654dbd4fcd-4jppl
  24 booking-service-654dbd4fcd-4qcjv
  35 booking-service-654dbd4fcd-7z8gl
  15 booking-service-654dbd4fcd-zlhxd
failed requests: 0   (every line is in /tmp/tatkal-scaling-watch.log)

# exit status 0
[10:08:40] tatkal-booking-microservices $
```

Kubernetes: one pod deleted and replaced, scaled to 3, 80 requests and none failed

```
[10:08:50] tatkal-booking-microservices $ record 4-kubernetes/rollout 'bash deploy/k8s/demo-rollout.sh'
$ bash deploy/k8s/demo-rollout.sh
# 2026-10-29 10:08:54 IST (UTC 04:38:54), git b118ea2

== booking-service 1.1: same code, APP_VERSION=1.1.0, loaded into minikube
 Image tatkal/inventory-service:1.1 Building
 Image tatkal/mock-bank-gateway:1.1 Building
 Image tatkal/payment-service:1.1 Building
 Image tatkal/notification-service:1.1 Building
 Image tatkal/booking-service:1.1 Building
 Image tatkal/booking-service:1.1 Built

== rolling update to 1.1
deployment.apps/booking-service image updated
Waiting for deployment "booking-service" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "booking-service" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "booking-service" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "booking-service" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "booking-service" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "booking-service" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "booking-service" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "booking-service" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "booking-service" rollout to finish: 1 old replicas are pending termination...
deployment "booking-service" successfully rolled out
POD                               IMAGE                          READY
booking-service-654dbd4fcd-7z8gl  tatkal/booking-service:1.0   true
booking-service-654dbd4fcd-zlhxd  tatkal/booking-service:1.0   true
booking-service-789bf6b667-5dd2m  tatkal/booking-service:1.1   true
booking-service-789bf6b667-j7l5t  tatkal/booking-service:1.1   true
booking-service-789bf6b667-p4kwv  tatkal/booking-service:1.1   true

== rollback to the previous version
Warning: resource deployments/booking-service was previously managed with 'kubectl apply'. Rolling back will not update the kubectl.kubernetes.io/last-applied-configuration a
nnotation, which may cause unexpected behavior on future 'kubectl apply' operations. Consider using 'kubectl apply' with your previous configuration file instead.
deployment.apps/booking-service rolled back
Waiting for deployment "booking-service" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "booking-service" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "booking-service" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "booking-service" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "booking-service" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "booking-service" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "booking-service" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "booking-service" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "booking-service" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "booking-service" rollout to finish: 1 old replicas are pending termination...
deployment "booking-service" successfully rolled out
deployment.apps/booking-service
REVISION  CHANGE-CAUSE
2         booking-service 1.1.0
3         <none>

POD                               IMAGE                          READY
booking-service-654dbd4fcd-5vqck  tatkal/booking-service:1.0   true
booking-service-654dbd4fcd-bqmt9  tatkal/booking-service:1.0   true
booking-service-654dbd4fcd-sz9fd  tatkal/booking-service:1.0   true
booking-service-789bf6b667-j7l5t  tatkal/booking-service:1.1   true
booking-service-789bf6b667-p4kwv  tatkal/booking-service:1.1   true

== a release that cannot start: there is no image tatkal/booking-service:9.9
deployment.apps/booking-service image updated
Waiting for deployment "booking-service" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "booking-service" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "booking-service" rollout to finish: 1 out of 3 new replicas have been updated...
error: timed out waiting for the condition
(the rollout does not finish, the old pods keep serving)
POD                               IMAGE                          READY
booking-service-56645c555f-z2bgl  tatkal/booking-service:9.9   false
booking-service-654dbd4fcd-5vqck  tatkal/booking-service:1.0   true
booking-service-654dbd4fcd-bqmt9  tatkal/booking-service:1.0   true
booking-service-654dbd4fcd-sz9fd  tatkal/booking-service:1.0   true

== rollback
Warning: resource deployments/booking-service was previously managed with 'kubectl apply'. Rolling back will not update the kubectl.kubernetes.io/last-applied-configuration a
nnotation, which may cause unexpected behavior on future 'kubectl apply' operations. Consider using 'kubectl apply' with your previous configuration file instead.
deployment.apps/booking-service rolled back
deployment "booking-service" successfully rolled out
POD                               IMAGE                          READY
booking-service-56645c555f-z2bgl  tatkal/booking-service:9.9   false
booking-service-654dbd4fcd-5vqck  tatkal/booking-service:1.0   true
booking-service-654dbd4fcd-bqmt9  tatkal/booking-service:1.0   true
booking-service-654dbd4fcd-sz9fd  tatkal/booking-service:1.0   true

== what the client inside the cluster saw: 159 requests, versions in order
  26 1.0.0
   2 1.1.0
   2 1.0.0
   1 1.1.0
   1 1.0.0
  18 1.1.0
   1 1.0.0
   6 1.1.0
 102 1.0.0
failed requests: 0   (every line is in /tmp/tatkal-rollout-watch.log)

# exit status 0
[10:10:23] tatkal-booking-microservices $
```

Kubernetes: rolling update, rollback, a broken release and a second rollback, 159 requests and none failed